

DengueRadar

Full concept reference - DAISI Challenge Round 1 (Problem A2: Real-time outbreak forecasting)
Compiled 30 Sep 2026 | Team: Braedon (lead, platform & ML), Jingyi (weather data), Ziqi (dengue & places data), Komal (story & pitch)

This is a working reference, not the pitch deck. It consolidates the challenge brief, why the team changed direction on 28 Sep, the current solution design, and the decisions locked during the Round 1 judging-fit review on 30 Sep. Where a design element changed during that review, this document reflects the **current**, **locked** version - not the older wording still sitting in some working docs pending edit.

1. The Challenge

The Databricks AI Social Impact (DAISI) Challenge invites student teams from Singapore IHLs to build data-driven solutions to local social challenges using Databricks and open data from data.gov.sg. Round 1 is an idea submission - a 3-slide PDF via Devpost, due 6 Oct 2026 (team target: 4 Oct) - no working build required. Ten teams are shortlisted from roughly 140 entrants; three are selected at Demo Day (27 Oct).

Official brief (Problem A2, kept verbatim)

"Dengue is hyperendemic in Singapore, with outbreaks affecting tens of thousands of residents. Rising temperatures and erratic rainfall extend the Aedes breeding season. NEA cluster monitoring today is largely reactive; predictive risk intelligence could let communities act 2-4 weeks ahead." What to build: a near-real-time pipeline combining live dengue cluster data, weather station readings and geospatial data into a forward-looking dengue risk forecast by planning area, with streaming or incremental ingestion, weather feature engineering, and a predictive model on an interactive map.

Source: project-context.md, quoting the DAISI A2 problem statement.

Round 1 judging weights

Criterion	Weight	What it rewards
Problem fit & social impact	30%	A clear problem and describable benefit
Solution quality & originality	30%	A thoughtful solution, not a generic dashboard
Data & technical feasibility	25%	Named open datasets, credible Databricks architecture, feasible in two weeks
Clarity of submission	15%	Concise, well-structured concept note or deck

2. Why We Changed Direction (28 Sep 2026)

The team's original idea narrowed the product to senior vulnerability and Active Ageing Centre staff, with only the national level backed by a trained model. That didn't match the brief, which asks for a forward-looking forecast **by planning area**, near-real-time ingestion, and a predictive model on a map. The team ran a correct-course review and pivoted.

What changed

Area	Decision
Problem	NEA's public cluster dashboard is reactive - it shows clusters after people are already sick. Nobody shows residents and community leaders where risk is heading.
Model	Level 2 (the product): a trained 2-week risk forecast per planning area (LightGBM in MLflow). Level 1 (context): a national weekly forecast. An explainable points score is kept as a fallback baseline.
Users	Residents (a personal action list) and community leaders (town councils, RCs, condo MCSTs). NEA is the partner who acts on some causes. Seniors: a secondary "extra care" flag, not the headline.
Prescriptive	Each forecast shows its top drivers and maps them to actions with an owner and timing. No invented effect sizes.
Ingestion	Daily Lakeflow archive of NEA clusters (live since 28 Sep) plus incremental loading of 5-minute weather.

Source: [_bmad-output/planning-artifacts/sprint-change-proposal-2026-09-28.md](#); [docs/team-plan.md](#).

3. The Solution Design

DengueRadar predicts, then prescribes. **Level 1 (context)** forecasts national weekly dengue cases two weeks ahead. **Level 2 (the product)** forecasts each planning area's dengue cluster risk two weeks ahead with a trained model, explains the top drivers, and maps each driver to an action for residents and community leaders.

Source: [docs/model-definitions.md](#).

The three hypotheses

ID	Hypothesis	Features
H1 Weather lag	Warm periods, and rain 2-4 weeks earlier followed by dry spells, raise next-fortnight cluster activity. The monthly chart supports temperature (+0.29 at 4 months) but not rain alone.	hot_days_31c, rain_lag2..4, temp_lag, humidity_lag, flushing_days
H2 Spatial spillover	Clusters in neighbouring planning areas raise an area's own risk.	cluster_cases_adjacent, dist_to_nearest_cluster_km
H3 Recurrence	Areas with clusters in the past year recur, especially with older housing stock.	cluster_weeks_last_52, median_block_age

Targets, baselines and models

Name	Rule
B1 Persistence (main baseline)	Next 2 weeks = the last 4 weeks' average cluster cases in that area ('clusters carry on').
B2 Seasonal average	Same week averaged over training years.
B3 Explainable points score	Hand-set points: cluster size band, high-Aedes zone, national Rising, recent rain, hot days. Runs today, needs no trained model.
M1 Trained model	LightGBM (regression + Rising classifier) logged in MLflow, with a Poisson/logistic cross-check. If it doesn't beat B1/B2, the team reports that honestly.

Target caveat: the training archive holds only cases inside NEA clusters. The model predicts *cluster activity* (new or growing cluster cases), not total dengue cases - this must appear on any slide using it.

Validation methodology (the Round 1 proof point - see Section 5)

- **Time-ordered, never random.** Rolling-origin: train 2015-2018 / test 2019; train 2015-2019 / test Jan-Jul 2020. Never shuffle weeks.
- **Leave-areas-out check** (GroupKFold by planning area) - confirms the model learns weather and spillover patterns, not each area's own history.
- **No leakage.** Features for week t use only data available before week t; NEA publishes cases about a week late, so lags are 1+ weeks. Imputers/scalers/encoders are fit inside each fold only.
- **Cost-matrix threshold.** Missing a rising area (false negative) is weighted 5x an unnecessary alert (false positive) as a starting assumption. The threshold minimising expected cost is chosen, then capped so at most 15% of areas are High in a typical week.
- **Known weakness, stated up front:** published Singapore dengue models catch the timing of peaks but underpredict their size.

Method: [github/08-datathon-handbook/03-modeling/cross-validation-guide.md](#).

4. The Prescriptive Action Layer - Current Design

This section reflects the design **as locked** during the 30 Sep Round 1 judging-fit review. It supersedes the driver-to-action wording still sitting in docs/model-definitions.md §6 and the sprint-change-proposal, which have not yet been edited to match.

Two independent routing axes

Axis 1 - why it's rising. The top 3 SHAP (or B3 rule-based) drivers map to **up to 3 deduplicated, ranked actions** - not collapsed to a single "leading driver." Overlapping actions from different drivers collapse into one line rather than repeat.

Axis 2 - who caused it. NEA's Dengue Clusters feed tags breeding source per cluster: HOMES, PUBLIC_PLACES, CONSTRUCTION_SITES. This is a bonus layer, not the backbone, because the field is sparse (2 of 11 clusters populated as of the 27 Sep check).

Breeding-source tag	Routed action
CONSTRUCTION_SITES	NEA/BCA enforcement referral. This is real institutional routing - NEA/BCA jurisdiction is national, not tied to a planning-area boundary, so it does not depend on data the team doesn't have.
HOMES / PUBLIC_PLACES	Feeds the resident/community-leader checklist below, block-level where HDB data allows.
Tag absent (~80% of clusters)	Falls back to driver-only actions (Axis 1). The Area view states "root cause not yet identified by NEA" explicitly, rather than omitting it silently.

LOCKED DECISION

No institutional routing to a specific Town Council, RC, or MCST. URA planning areas do not align with Town Council / electoral / MCST boundaries, and no open dataset maps one to the other. Every community-leader action is instead framed as a checklist that **any engaged resident, RC volunteer, or MCST council member in a flagged area can read and act on or escalate themselves**.

Why this matters: naming a specific institution implies a routing capability the data can't support - a feasibility gap that would read as a solution-quality weakness to a judge who checks it.

Example action table (illustrative - CONFIRM final wording against NEA's public prevention advice)

Leading driver	Resident action (this week)	Timing
Recent rain then warm spell (H1)	Empty flower-pot plates and containers, clear gully traps, cover storage	Within 7 days
Active cluster nearby (H2)	Repellent, long sleeves at dawn/dusk, see a doctor early for fever	Within 3 days
Recurrence, older blocks (H3)	Routine weekly checks	Within 14 days

Extra-care flag: where an area's 65+ share is in the top third, add "check on older neighbours and family members" - the only place seniors enter the product.

LOCKED DECISION

The action layer is deliberately deterministic and rule-based, not model-generated free text. This is a stated design principle, not an admission of simplicity: health guidance that real people act on needs to be auditable and NEA-sourced, not generated by an unverified model.

The forecasting model can be complex because getting it wrong only costs accuracy; the action layer is kept simple because getting it wrong costs someone acting on bad advice.

5. Round 1 Judging-Fit Review - What Changed (30 Sep 2026)

A forge session pressure-tested the Round 1 submission against the rubric in Section 1, aiming specifically at standing out among roughly 140 competing teams. Full session trace:

`_bmad-output/forge/dengueradar-round1-judging-fit/.memlog.md`.

LOCKED DECISION

The submission does not depend on a model accuracy number anywhere. No task exists before the 4 Oct deadline to train the model, run SHAP, or compute a backtest - that work is a Build Sprint deliverable (12-26 Oct), gated on being shortlisted. Round 1 requires no code; the feasibility criterion asks for credible architecture, not a finished result.

Slide 3's proof point is therefore the validation methodology in Section 3, not a fabricated or pre-empted number. This also satisfies the team's own rule: never claim a lead time before the backtest measures it.

LOCKED DECISION

The idea-originality claim is not the forecast or the map - the brief asks every one of ~140 teams for exactly that - and not SHAP/explainability, a standard technique. The real differentiator: official Singapore open data has no historical dengue record below the national level, a wall any competent team building a genuinely trained per-area model will hit. DengueRadar's original contribution is identifying and validating two independent, citable archives - SGCharts (256 snapshots, 2015-2020) and the Nature Scientific Data cluster CSV (2022, 213 subzones mapped) - to manufacture the per-area training history the brief requires.

Claimed precisely as "we found and solved the constraint with a reproducible method," not "nobody else could find this" - other capable teams may independently reach the same archives.

Two factual fixes, resolved (not just flagged)

Item	Resolution
2020 dengue case count	35,261 - independently reproduced by summing analysis/data/weekly_idb.csv (Dengue Fever rows, 2020-W01..W53). Matches DATASETS.md. The competing 35,012 figure in docs/checkpoint-1oct.md has no traceable source and should be dropped.
"Rising temperature and erratic rainfall extend the Aedes season" (slide 1, why-now)	Cite directly to the official DAISI A2 brief - the cheapest possible source. Caution: the team's own research (<code>_bmad-output/planning-artifacts/research/technical-dengueradar-feasibility-2026-09-28</code>) found this is more nuanced - temperature above 31C is often dampening, and heavy "flushing" rain lowers risk. H1 already reflects the nuanced version; don't let a naive "hot+rainy=worse" framing contradict it elsewhere.

Three pillars - where the submission stands

Pillar	Status
Idea originality	Fixed by the data-wall reframe above.
Solution soundness	Strong: validation methodology, honest sparsity handling, no-institutional-routing honesty, auditable-by-design actions.
Execution credibility	Strong evidence exists (live Lakeflow archive since 28 Sep, Free Edition capability check passed, real NEA field data already explored) but how it lands on the slide is still open.

Still open

- Execution-credibility pillar: compressing the "already-running pipeline" evidence onto a slide.
- Doc edits needed to match these locks: docs/model-definitions.md §6, the sprint-change-proposal, and docs/checkpoint-1oct.md still carry superseded wording.

6. Core Datasets at a Glance

#	Dataset	Agency	Tier	Used for
1	Weekly Infectious Disease Bulletin	MOH	Core	National model target (weekly dengue cases)
2	Dengue Clusters (incl. breeding-source tags)	NEA	Core	Local exposure points, cause-routing, live model input
3	Areas with High Aedes Population	NEA	Core	Mosquito hotspots
5/6	Master Plan 2019 Subzone / Planning Area Boundaries	URA	Core	Scoring unit and map
7-10	Historical & real-time rainfall, temperature, humidity	NEA	Core	Weather features, incremental ingestion
4	Residents by Planning Area/Subzone, Age, Sex (Census 2020)	SingStat	Should	Population denominator, 65+ extra-care flag

Training-only archives (cited, offline): SGCharts cluster archive (256 snapshots, 3 Jul 2015 - 6 Nov 2020) and the Nature Scientific Data cluster CSV (20 weekly snapshots, 28 Feb - 9 Jul 2020, 213 subzones mapped, DOI 10.1038/s41597-022-01666-y, CC BY 4.0).

Full list of 18 datasets with links, formats and owners: [DATASETS.md](#).

7. Team & Round 1 Schedule

Person	Role	Round 1 focus
Braedon	Team lead, Platform & Data Architect, ML Modeler	Workspace/Free Edition tests, architecture diagram, lag chart, model design, prescriptive rules
Jingyi	Data: weather	Verify/download weather data, map stations to planning areas
Ziqi	Data: dengue & places	Verify dengue/boundary/population data, build the per-area training panel
Komal	Story & Pitch Lead	Sourced statistics, resident story, mock-ups, 3 slides, Devpost submission

Date	Milestone
1 Oct	Checkpoint: does the archive give a usable per-area signal? Adjust now if not.
2-3 Oct	Draft the 3 slides from the official template.
3 Oct	Mock judging: scores plus the hardest questions.
4 Oct	Final polish, export PDF, submit on Devpost.
6 Oct, 11:59pm SGT	Deadline (5-6 Oct held as buffer).

8. Open Risks & Questions

- Lead time not yet measured - never claim a specific number before the backtest runs (Sprint).
- Thin per-area training window: weekly data to May 2020, monthly after; target is cluster activity only, not total cases.

- Streaming/Auto Loader on Free Edition: tested but not yet fully confirmed for the incremental-ingestion path.
- NEA-dashboard wording and action wording still need confirmation against NEA's own public pages before any slide uses them.
- Nobody has yet spoken to a town council, RC, or NEA contact to sanity-check the actions.
- The team's own archive will be roughly 30 days deep by Demo Day.

Full session traces and prior (superseded) design iterations are kept for history in `_bmad-output/forge/` and `_bmad-output/planning-artifacts/`, each marked with its superseded status.